

On the Quantum Hardness of Solving Isomorphism Problems as Nonabelian Hidden Shift Problems

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Organization of the talk

Graph isomorphism

Hidden **subgroup** approach

Hidden **shift** approach

- Definition of nonabelian H-Shift-P
- Natural quantum approach to solving H-Shift-P
- Isomorphism problems as H-Shift-P
- Structure of hidden shift states
- Single register measurement do not suffice
- Lower bound on the number of copies

Graph isomorphism

Γ graph on n vertices and $g \in S_n$ permutation

S_n acts on the set of n -vertex graphs: $g(\Gamma)$ is the graph obtained from Γ by permuting the vertices according to g

Γ rigid $\Leftrightarrow \nexists g \in S_n : g(\Gamma) = \Gamma$

Γ_0 and Γ_1 isomorphic $\Leftrightarrow \exists s \in S_n : s(\Gamma_1) = \Gamma_0$

s is the isomorphism between Γ_1 and Γ_0

Problem: given two rigid graphs decide if they are isomorphic or not

Hidden subgroup problem

$f : G \rightarrow S$ where G is a group and S a finite set

f hides a subgroup H , that is,

- f is constant on the cosets G/H , and
- f is different on distinct cosets

Problem: determine H

Graph isomorphism as HSP

Quantum approach to solving HSP

Hardness of GI as HSP

Definition of nonabelian H-Shift-P

black-box access to two injective functions

$f_0 : G \rightarrow S$ and $f_1 : G \rightarrow S$,

where G is a (nonabelian) group and S is a finite set

the functions f_0 and f_1 are promised to satisfy:

(1) there is an $s \in G$ such that $f_0(g) = f_1(gs)$ for all $g \in G$

s is the hidden shift

(2) the images of f_0 and f_1 are disjoint

no hidden shift

Problem: determine if (1) or (2)

Graph isomorphism as H-Shift-P

Γ graph on n vertices and $g \in S_n$ permutation

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Γ rigid $\Leftrightarrow \nexists g \in S_n : g(\Gamma) = \Gamma$

Γ_0 and Γ_1 isomorphic $\Leftrightarrow \exists s \in S_n : s(\Gamma_1) = \Gamma_0$

s is the isomorphism between Γ_1 and Γ_0

Graph isomorphism as H-Shift-P

let Γ_0 and Γ_1 be two rigid n -vertex graphs

define $f_0(g) = g(\Gamma_0)$ and $f_1(g) = g(\Gamma_1)$

clearly, f_0 and f_1 are injective because both graphs rigid

(1) assume $s(\Gamma_1) = \Gamma_0$ for $s \in S_n \quad \Rightarrow \quad f_0(g) = f_1(gs)$

(2) assume Γ_1 and Γ_0 are not isomorphic $\Rightarrow$

the images of f_0 and f_1 are disjoint

Quantum approach to solving H-Shift-P

prepare a uniform superposition over $i \in \mathbb{Z}_2$ and $g \in G$,
compute the value of $f_i(g)$, and measure the third register

$$\frac{1}{\sqrt{2|G|}} \sum_{g \in G} (|0, g, f_0(g)\rangle + |1, g, f_1(g)\rangle)$$

- if there is a hidden shift s , then we are left with

$$|\phi_{s,g}\rangle := \frac{1}{\sqrt{2}} (|0, g\rangle + |1, gs\rangle) \in \mathbb{C}^2 \otimes \mathbb{C}^{|G|}$$

for some uniformly random (unknown) $g \in G$

- if there is no hidden shift, then we obtain

$$|i, g\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^{|G|}$$

for uniformly random $i \in \mathbb{Z}_2$ and $g \in G$

Hidden shift s

if s is the hidden shift, then the density matrix is

$$\gamma_1(s) := \frac{1}{|G|} \sum_{g \in G} |\phi_{s,g}\rangle \langle \phi_{s,g}|$$

Observation: $\gamma_1(s)$ can be expressed as R

$$\gamma_1(s) = \frac{1}{2|G|} \begin{pmatrix} I & R(s) \\ R(s^{-1}) & I \end{pmatrix}$$

where R is the right regular representation

$$R(s)|g\rangle = |gs^{-1}\rangle$$

No hidden shift

if there is no hidden shift, the density matrix is

$$\gamma_2 := \frac{1}{2|G|} \sum_{i \in \mathbb{Z}_2} \sum_{g \in G} |i, g\rangle \langle i, g|$$

γ_2 is the maximally mixed state

$$\gamma_2 := \frac{1}{2|G|} \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}$$

H-Shift-P as state discrimination

we can apply the above procedure k times to obtain k copies of the hidden shift state (or the maximally mixed state)

assume that in the case where there is a hidden shift s , it is equally likely to correspond to any element of G

then the problem is to distinguish the two density operators

$$\gamma_1^{(k)} := \frac{1}{|G|} \sum_{s \in G} \gamma_1^{(k)}(s)$$

$$\gamma_2^{(k)} := \frac{1}{(2|G|)^2} I$$

where $\gamma_1^{(k)}(s) := \gamma_1(s)^{\otimes k}$

Structure of the hidden shift states

Recall

$$\gamma_1(s) = \frac{1}{2|G|} \begin{pmatrix} I & R(s) \\ R(s^{-1}) & I \end{pmatrix}$$

where R the right regular representation $R(s)|g\rangle = |gs^{-1}\rangle$

$\hat{G}$ a set of all irreducible (inequivalent) representations of G

F the (corresponding) Fourier transform over G

then we have block decomposition

$$FR(s)F^\dagger = \bigoplus_{\rho \in \hat{G}} I_{d_\rho} \otimes \rho(s)$$

Structure of the hidden shift states

in the Fourier basis we have the block decomposition

$$\begin{aligned}\tilde{\gamma}_1(s) &= \frac{1}{2|G|} \bigoplus_{\rho \in \hat{G}} I_{d_\rho} \otimes \underbrace{\begin{pmatrix} I_{d_\rho} & \rho(s) \\ \rho(s^{-1}) & I_{d_\rho} \end{pmatrix}}_{B^\rho(s)} \\ \tilde{\gamma}_2 &:= \frac{1}{2|G|} \bigoplus_{\rho \in \hat{G}} I_{d_\rho} \otimes \underbrace{\begin{pmatrix} I_{d_\rho} & 0 \\ 0 & I_{d_\rho} \end{pmatrix}}_{I_{2d_\rho}}\end{aligned}$$

Single-register measurements

without loss of generality any POVM used to distinguish among these states has the same block structure



first measure representation name (weak Fourier sampling),
then perform arbitrary POVM inside the block

the various irreducible representations ρ occur independently according to the Plancherel distribution

$$\Pr(\rho) = \frac{d_\rho^2}{|G|}$$

regardless of whether or not there is a hidden shift

Single register measurements

$\mathcal{E} = \{a_1|\psi_1\rangle\langle\psi_1|, \dots, a_r|\psi_r\rangle\langle\psi_r|\}$ arbitrary POVM acting on subspace of dimension $2d_\rho$ corresponding to observed ρ

no hidden shift $\rightarrow$

post-measurement state is $I_{2d_\rho}/2d_\rho$ and probability of j is

$$p_2(j) = \frac{a_j}{2d_\rho} \langle\psi_j|I_{2d_\rho}|\psi_j\rangle = \frac{a_j}{2d_\rho}$$

hidden shift $s \rightarrow$

post-measurement state is $B^\rho(s)/2d_\rho$ and probability of j is

$$p_1(j|s) = \frac{a_j}{2d_\rho} \langle\psi_j|B^\rho(s)|\psi_j\rangle$$

let P_2 and $P_{1,s}$ corresponding probability distributions

Single register measurements

single register measurement do not suffice because

$$(*) \quad \Pr_{s \in G, \rho \in \hat{G}}(\|P_{1,s} - P_2\|_{\text{TV}} \geq e^{-\Theta(n)}) \leq e^{-\Theta(n)}$$

define $p_1(j) := \frac{1}{|G|} \sum_{s \in G} p_1(j|s)$

let P_1 be the corresponding probability distribution

we prove $(*)$ by showing

- P_1 and P_2 are equal for all $\rho \in \hat{G} - \{\hat{1}\}$
- $\Pr_{s \in G, \rho \in \hat{G}}(\|P_{1,s} - P_1\|_{\text{TV}} \geq e^{-\Theta(n)}) \leq e^{-\Theta(n)}$

Single register measurements

let $\rho \in \hat{G}$ be fixed and d_ρ be its degree

let σ_j be the variance of the random variable $p_1(j|s)$ with respect to $s \in G$ uniformly at random

$$\sigma_j^2 := \frac{1}{|G|} \sum_{s \in G} p_1(j|s)^2 - p_1(j)^2$$

Chebyshev's inequality

$$\Pr_{s \in G}(|p_1(j|s) - p_1(j)| \geq a_j c) \leq \frac{\sigma_j^2}{a_j^2 c^2}$$

define

$$J_{\text{bad}}^s := \{j \in J : |p_1(j|s) - p_1(j)| \geq a_j c\} \text{ and } J_{\text{good}}^s := J - J_{\text{bad}}^s$$

Single register measurements

Now we can write

$$\|P_{1,s} - P_1\|_{\text{TV}} = \sum_{j \in J_{\text{good}}^s} |p_1(j|s) - p_1(j)| + \sum_{j \in J_{\text{bad}}^s} |p_1(j|s) - p_1(j)|$$

Markov's inequality gives the bound

$$\Pr\left(\sum_{j \in J_{\text{bad}}^s} a_j \geq c'\right) \leq \frac{1}{d_\rho^2 c^2 c'}$$

Lower number of copies